UNIT 2 · FORCES & NEWTON'S LAWS · WEEK 1

Problem Set 7 — FBDs & Newton's Laws

Physics 2026–27 · due at the start of class, Tuesday, November 10

NAME _____ DATE _____

How to work this set. Every force problem starts with a free-body diagram: isolate the object as a dot, draw every force acting *on* it (nothing it exerts on anything else), label each arrow with its symbol. Then KDEA with $\Sigma F = ma$ — or $\Sigma F = 0$ if the velocity isn't changing. Force unit: $N = kg \cdot m/s^2$, exactly as Unit 0 promised.

1 Three FBDs

DIAGRAMS

Draw a labeled free-body diagram for each. Arrow lengths should reflect relative magnitudes.

a. A textbook resting on a level table.

b. A shop sign hanging motionless from a single vertical rope.

c. A shopping cart rolling at *constant velocity* across a flat, frictionless floor — nobody touching it. (Careful: how many forces does it actually have?)

DIAGRAMS

A

B

C

2 Equilibrium or not?

FIRST LAW

For each object, state whether $\Sigma F = 0$, and how you know:

a. A parked bus.

b. A hockey puck sliding across frictionless ice at a constant 4 m/s.

c. A Metrorail train pulling away from the station, speeding up.

d. One sentence: what belief about moving objects does (b) disprove?

ANSWERS

3 Straight F = ma

SECOND LAW

a. A $2.5\ kg$ lab cart feels a net force of $10\ N$. Find its acceleration.

b. A $1400\ kg$ car accelerates at $2.5\ m/s^2$. What net force is required?

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	ΣF = MA
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Tug of war

NET FORCE

A 6.0 kg crate sits on a frictionless floor. One student pushes with 25 N to the right, another with 10 N to the left. Draw the FBD, then run KDEA for the acceleration — magnitude *and* direction.

FBD

K	EVERY GIVEN, WITH UNITS · PICK A POSITIVE DIRECTION
D	THE TARGET & ITS UNIT
E	ΣF WITH SIGNS, THEN = MA
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Forces at a right angle

VECTORS RETURN

A 12 kg crate sits on frictionless ice. One person pushes east with 50 N; another pushes north with 120 N at the same time. Workshop III trig, force edition:

a. Find the magnitude of the net force.

b. Find its direction (angle north of east).

c. Find the crate's acceleration — magnitude and direction.

K	EVERY GIVEN, WITH UNITS
D	THREE TARGETS, THREE UNITS
E	PYTHAGOREAN + TAN⁻¹, THEN ΣF = MA
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The normal force is not a constant

CONCEPT

A $1.2\ \text{kg}$ book rests on a table.

a. Find N. What equation actually justifies it — and what condition makes that equation true?

b. Now you press straight down on the book with an extra $5.0\ \text{N}$. What is N now?

c. One sentence: why is "N = mg" a special case rather than a law?

WORK

Graded on the homework 4–3–2–1 scale. An FBD with a phantom "motion force" on it is the week's classic error — constant velocity needs no force at all.